Finite Element Analysis of Truss with Iterative Area Optimization

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***Abstract*—This report presents a finite element analysis (FEA) of a two-dimensional truss structure that was done using Abaqus software to determine member stresses and nodal displacements, followed by a fully stressed design optimization. Results from the initial analysis revealed that element stresses ranged from –431 psi to 345 psi when all cross-sectional areas were equal to 3.48 in^2. Additionally, the displacement results confirmed correct boundary condition and load implementation through near-zero displacements at constrained nodes and higher displacements at nodes with forces applied. The fully stressed design procedure then iteratively adjusted the cross-sectional areas of each member to bring element stresses closer to the allowable limit of 350 psi while adhering to a minimum area constraint of 0.1 in^2. After four iterations, most members achieved stresses near the allowable value, demonstrating convergence and the effectiveness of the design approach. The study validated that FEA can efficiently optimize truss performance by balancing material use, stress distribution, and design constraints.**

***Index Terms*—Cross Sectional Area, Displacement, Stress, Truss**

I. Problem Statement

THE objective of this project was to analyze a two-dimensional truss structure using the finite element method and to perform a fully stressed design optimization based on the obtained stress results. The truss geometry, loading conditions, and boundary conditions were provided in the project handout and are shown in Fig. 1 below where nodes and element numbers are labeled in red and purple, respectively.

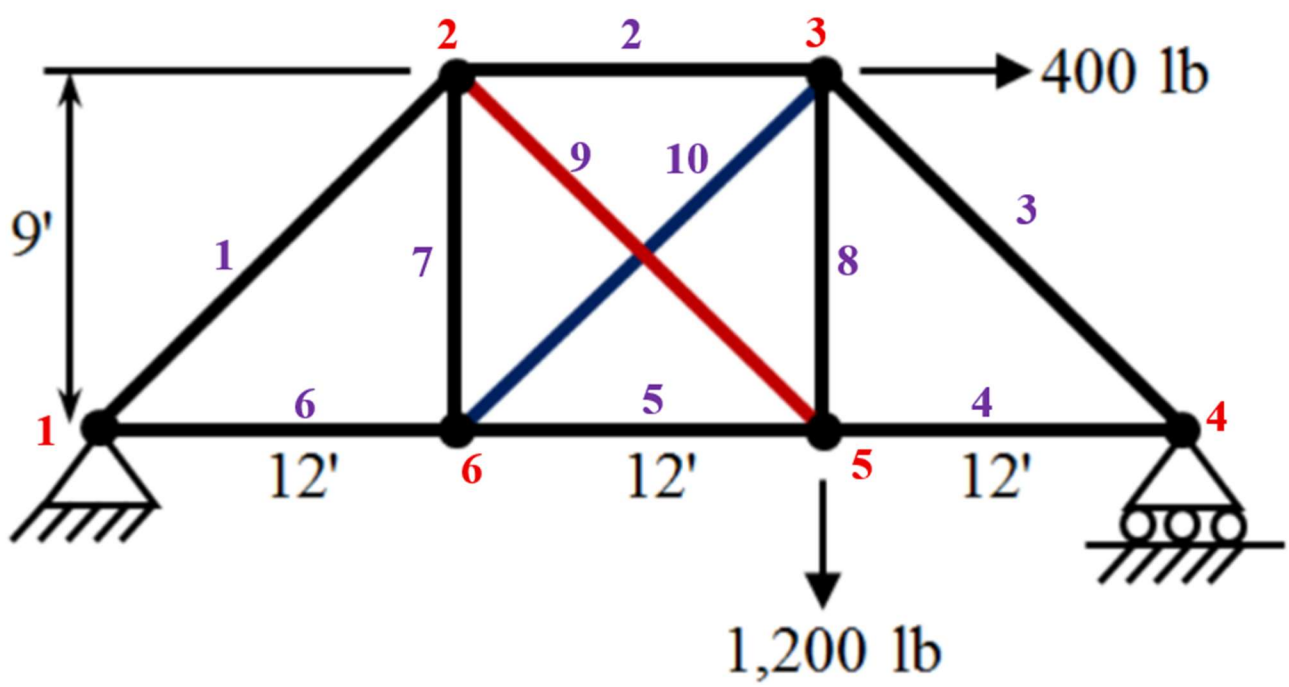

Fig. 1. Truss geometry from problem statement with a pin, roller, horizontal force, vertical force and labeled nodes and elements.

The material properties provided were a Young's modulus, *E,* of 10,000 psi and Poisson's ratio, *v,* of 0.3. The initial cross-

sectional area, *A,* of each truss element was equal to 3.xx where xx was the last two digits of a UFID, and in this case area was equal to 3.48 in^2.

The first portion of the analysis (part a) involved determining the normal stress in the axial direction, σ_{xx}, in each member and the nodal displacements in both the horizontal and vertical directions, denoted as $u$and v, respectively. The truss was modeled on Abaqus, an FEA software suite, to find the results for this part of the project.

In the second portion, a fully stressed design procedure was implemented to optimize the cross-sectional areas of the truss members. The goal was to adjust the area of each member so that all members would fall within the allowable stress limit. The allowable stress, $\sigma_{\text{allowable}}$, for this problem was equal to 350 psi and the minimum permissible cross-sectional area was 0.1 in^2. The new area of each member was calculated using (1) below where $\sigma_{old}^{(e)}$ was the old stress value for element e, $\sigma_{\text{allowable}}$ was the allowable stress equal to 350 psi, $A_{old}^{(e)}$ was the old area of element e, and $A_{new}^{(e)}$ was the resulting new area of element e.

$$A_{new}^{(e)} = \frac{\sigma_{old}^{(e)}}{\sigma_{allowable}} A_{old}^{(e)} \tag{1}$$

This process was iterated for four cycles since the provided truss was a statically indeterminate system, meaning that the number of unknown internal forces or reactions exceeded the number of available static equilibrium equations, and therefore analytical methods could not be used.

II. Procedure

Part a: Truss Analysis

The finite element model was created and analyzed using the Abaqus software. A new part was created, and the modeling space was set to 2D planar, the part type to deformable, and the base feature to wire. The truss geometry was then sketched by inputting the nodal coordinates derived from the dimensions provided in Fig. 1. The dimensions were converted from feet to inches, and the final node coordinates used were $(0,0)$, $(144,0)$, $(288,0)$, $(432,0)$, $(288,108)$, and $(144,108)$.

Next, the material properties were defined by navigating to the Material module and selecting “Mechanical”, “Elasticity” and “Elastic”. The given material property values were then added for Young’s modulus and Poisson’s ratio. A section with an initial cross-sectional area of 3.48 in^2 was created and assigned to all elements in the truss.

A static step was created to enable field output requests for later extraction of stress and displacement data. The boundary conditions were then applied as follows: Node 1 was assigned a pinned boundary condition, constraining both horizontal and vertical motion, while Node 4 was assigned a roller boundary condition, restricting only vertical displacement. The loading conditions consisted of a horizontal force of 400 lb applied at Node 3 and a vertical force of –1200 lb applied at Node 5.

The truss was then meshed by selecting the “Truss” element type, seeding each edge by number, and assigning one element per member. After seeding, the mesh was generated by selecting “Mesh Instance”. A job was created and submitted for analysis; however, the initial run produced inaccurate results because Abaqus automatically inserted an additional node at the element intersection. To correct this, the input file (.inp) was manually edited to remove the extra nodes and ensure proper element and nodal connectivity. The corrected input file was then resubmitted as a new job.

After a successful analysis, the element stress and nodal displacement results were exported by pressing the “Report” tab and then the “Field Output” option. For stresses, the S_{11} component at the integration point position was selected, and for displacements, the U field at unique nodal positions was extracted. These results were saved into a report file (.rpt) for further analysis.

Part b: Fully Stressed Design

Following completion of the initial truss analysis, the stresses and displacements recorded in the .rpt file were transferred to an Excel spreadsheet for post-processing. The initial cross-sectional area and the allowable stress ($\sigma_{allow} = 350$ psi) were entered into the sheet, and the new areas for each member were calculated using the fully stressed design equation (1). If any computed cross-sectional area fell below the minimum allowable value of 0.1 in^2, it was adjusted upward to satisfy the constraint.

In Abaqus, ten new sections were created—one for each truss member—with cross-sectional areas corresponding to the calculated values from the Excel sheet. A job was created based off the adjusted model, but the input (.inp) file had to be adjusted to ensure the different areas were assigned to the correct elements. Once the appropriate section assignments were verified, the modified input file was submitted, and the updated stresses were saved to the .rpt file in the same manner as in part a.

This process of recalculating areas, updating the inp file, and re-running the analysis was repeated for four total iterations. After the final iteration, the data for all stresses and corresponding cross-sectional areas were compiled to evaluate the convergence toward a fully stressed design.

III. Results

Part a: Results

In part a, stress and displacement data was saved for the truss structure with the cross sectional area of all elements set equal to 3.48 in^2. The initial undeformed truss is shown in Fig. 2 below revealing the shape of the structure before applying forces.

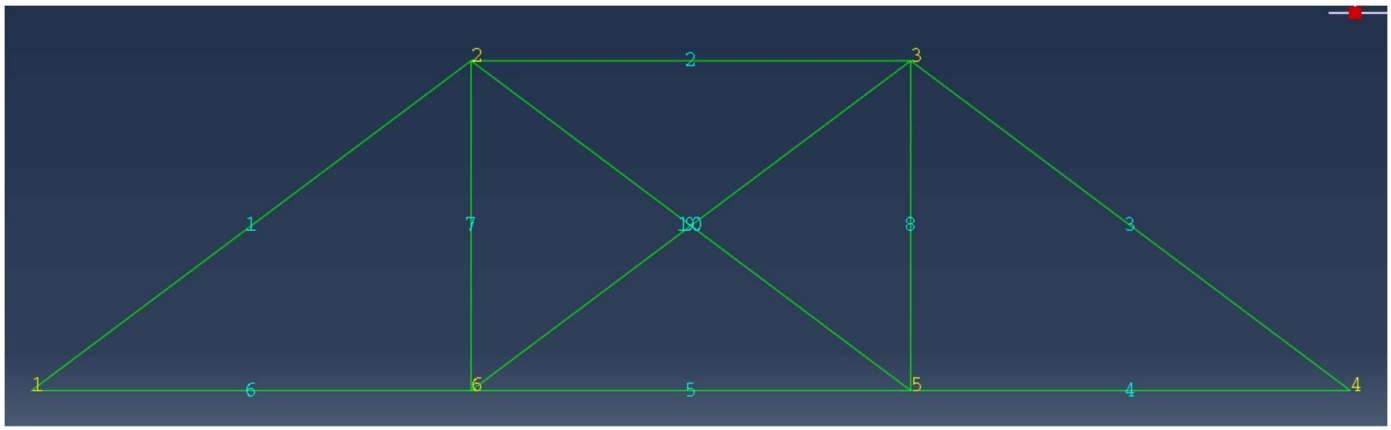

Fig. 2. Undeformed truss structure before applying forces with node and element labels.

After the concentrated forces were applied, the deformed results were displayed to reveal the physical displacement of nodes and the stress level (coded in color according to the legend) of elements as shown in Fig. 3 below.

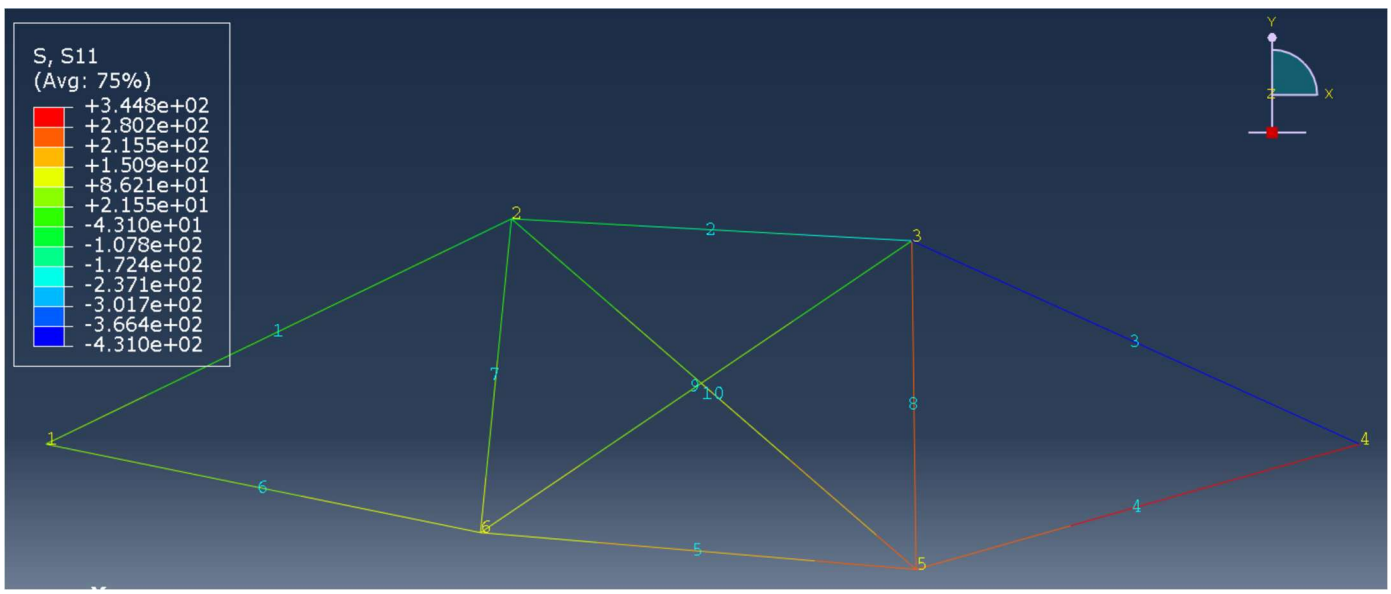

Fig. 3. Deformed truss structure after applying forces with node and element labels and stress legend.

The horizontal stresses, σ_{xx}, and the corresponding element numbers are detailed in Table I below for the initial test case with equal cross-sectional areas in each member.

Table I
Initial Case Element Stress Data

Element	Stress (σ_{xx}) [psi]
1	-143.678
2	-218.178
3	-431.034
4	344.828
5	241.592
6	229.885

Table I Continued
Initial Case Element Stress Data

Element	Stress (σ_{xx}) [psi]
7	8.78033
8	267.401
9	-14.6339
10	129.044

For the initial case with all equal cross-sectional areas, nodal displacement data was also saved for the deformed truss structure as shown in Table II below where *u* is the horizontal displacement and *v* is the vertical displacement. The magnitude of the total displacements is also included. Note that the displacements for the constrained directions of the nodes with boundary conditions (1 and 4) are essentially equal to zero.

Table II
Initial Case Displacement Data

Node	Magnitude [in]	*u* [in]	*v* [in]
1	0.000	$4.0e^{-34}$	$-3.0e^{-34}$
2	18.856	9.138	-16.494
3	21.464	5.996	-20.609
4	11.755	11.755	$-9.0e^{-34}$
5	24.458	6.789	-23.497
6	16.916	3.310	-16.589

Part b: Results

The next part of this project involved a fully stressed design procedure to optimize cross sectional areas and bring all element stresses within the allowable stress limit. Tables III, IV, V, and VI below show the results of the four iterations of calculating new element cross sectional areas, $A_{new}^{(e)}$, and the corresponding new stresses, $\sigma_{new}^{(e)}$, based off the old areas, $A_{old}^{(e)}$, and the old stresses, $\sigma_{old}^{(e)}$.

Table III
Fully Stressed Design Areas and Stresses of First Iteration

Trial 1				
Element	$A_{old}^{(e)}$ [in^2]	$\sigma_{old}^{(e)}$ [psi]	$A_{new}^{(e)}$ [in^2]	$\sigma_{new}^{(e)}$ [psi]
1	3.480	-143.678	1.429	-349.895
2	3.480	-218.178	2.169	-358.756
3	3.480	-431.034	4.286	-349.977
4	3.480	344.828	3.429	349.956
5	3.480	241.592	2.402	342.156
6	3.480	229.885	2.286	349.956
7	3.480	8.78033	0.100	188.428
8	3.480	267.401	2.659	344.638
9	3.480	-14.6339	0.146	-187.137
10	3.480	129.044	1.283	368.416

Table IV
Fully Stressed Design Areas and Stresses of Second Iteration

Trial 2				
Element	$A_{old}^{(e)}$ [in^2]	$\sigma_{old}^{(e)}$ [psi]	$A_{new}^{(e)}$ [in^2]	$\sigma_{new}^{(e)}$ [psi]
1	1.429	-349.895	1.428	-350.14
2	2.169	-358.756	2.224	-352.789
3	4.286	-349.977	4.285	-349.977
4	3.429	349.956	3.428	350.058
5	2.402	342.156	2.348	347.273
6	2.286	349.956	2.285	350.109
7	0.100	188.428	0.100	115.477
8	2.659	344.638	2.618	348.185
9	0.146	-187.137	0.100	-192.462
10	1.283	368.416	1.351	355.85

Table V
Fully Stressed Design Areas and Stresses of Third Iteration

Trial 3				
Element	$A_{old}^{(e)}$ [in^2]	$\sigma_{old}^{(e)}$ [psi]	$A_{new}^{(e)}$ [in^2]	$\sigma_{new}^{(e)}$ [psi]
1	1.428	-350.14	1.429	-349.895
2	2.224	-352.789	2.241	-350.374
3	4.285	-349.977	4.285	-350.058
4	3.428	350.058	3.429	349.956
5	2.348	347.273	2.330	349.705
6	2.285	350.109	2.286	349.956
7	0.100	115.477	0.100	111.093
8	2.618	348.185	2.604	349.888
9	0.100	-192.462	0.100	-185.155
10	1.351	355.85	1.373	350.681

Table VI
Fully Stressed Design Areas and Stresses of Fourth Iteration

Trial 4				
Element	$A_{old}^{(e)}$ [in^2]	$\sigma_{old}^{(e)}$ [psi]	$A_{new}^{(e)}$ [in^2]	$\sigma_{new}^{(e)}$ [psi]
1	1.429	-349.895	1.428	-350.14
2	2.241	-350.374	2.244	-349.939
3	4.285	-350.058	4.286	-349.977
4	3.429	349.956	3.428	350.058
5	2.330	349.705	2.328	349.973
6	2.286	349.956	2.286	349.956
7	0.100	111.093	0.100	110.531
8	2.604	349.888	2.604	349.867
9	0.100	-185.155	0.100	-184.219
10	1.373	350.681	1.376	349.984

IV. Discussion

Part a: Stress Analysis

The finite element analysis of the truss provided insight into the stress distribution and deformation behavior under the applied loading conditions. For the initial model, with all members having a cross-sectional area of 3.48 in², the computed element stresses ranged from approximately -431.034 psi (element 3, in compression) to 344.828 psi (element 4, in tension). The most highly stressed member, element 3, exceeded the allowable stress of 350 psi, indicating that certain members were over-stressed under the given loading configuration. Conversely, some elements such as element 7 and element 9 exhibited very low stress magnitudes of 8.780 psi and –14.634 psi, respectively, demonstrating that those members carried minimal load.

Part a: Displacement Analysis

The nodal displacement results also aligned with expectations based on the boundary conditions. At node 1, displacement in both the horizontal and vertical directions was approximately equal to zero due to the pin boundary condition that constrained both directions. Similarly, at node 4, displacement in the vertical direction was approximately equal to zero due to the roller boundary condition that constrained displacement in the vertical direction. This confirmed that the required boundary conditions from the problem statement were correctly applied. The largest displacement magnitude occurred at Node 5, with a resultant displacement magnitude of 24.458 in, corresponding to $u = 6.789$ in and $v = -23.497$ in. This node experienced the downward vertical load of –1200 lb, which explains the significant downward deflection observed in the deformed geometry.

Part b: Analysis

Following the initial analysis, the fully stressed design process was conducted to optimize the cross-sectional areas of each element so that the member stresses approached the allowable limit of 350 psi. After four iterations, the stresses converged towards the allowable stress with optimized areas, demonstrating that the method was effective. While majority of the elements were just below the allowable stress, two elements were slightly above this value, and two elements were far below this value. Elements 1 and 4 had final stress values equal to -350.14 psi and 350.058 psi, respectively, after the fourth iteration. Although they nearly

approached the allowable stress, they were still technically out of spec. This likely means that more iterations or more exact area values (more than three decimal places) would need to be used to bring these values within the allowable limit. On the other hand, elements 7 and 9 experienced a final stress equal to 111.093 psi and -185.155 psi, respectively, falling far below the allowable stress of 350 psi. This occurred due to the minimum cross-sectional area constraint of 0.1 in^2 which prevented these elements from reaching the allowable stress as they would have needed a far greater reduction in area to achieve that value.

Conclusion

Overall, the iterative process demonstrated the efficiency of the fully stressed design approach. Members that were initially over-stressed had their areas increased, while those that were under-stressed had their areas reduced until reaching the allowable stress limit or the minimum cross-sectional area. By the fourth iteration, the design achieved a near-uniform stress distribution across most members except for those that were limited by the minimum area constraint. The analysis also confirmed the correctness of the boundary condition setup and showed that the structural deformation pattern corresponded logically to the applied loads. This iterative FEA approach successfully produced a structurally efficient truss design where element cross-sectional area was optimized and stress levels were balanced throughout the structure.